

JACKSON MORRIS

CURRICULUM VITAE

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EMPLOYMENT

Vanderbilt University
Postdoctoral Scholar

Vanderbilt, TN
2027-2029

Simon Laufer Mathematical Sciences Institute
Postdoctoral Scholar

Berkeley, CA
Fall 2026

-Program: Motivic Homotopy Theory: Connections and Applications

EDUCATION

University of Washington
Ph.D. in Mathematics

Seattle, WA
2021-2026

-Advisors: Kyle Ormsby and John Palmieri

-Thesis title: Cooperations in motivic homotopy theory

M.S. in Mathematics

2021 - 2023

University of Kentucky

B.S. in Mathematics *magna cum laude*

Lexington, KY
2017-2020

RESEARCH INTERESTS

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, splittings, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

PUBLICATIONS AND PREPRINTS

10. *Derived Galois and fracture models for cellular motivic spectra over $\mathbb{Q}_2$* , joint with Luca De Paulis, Andrea Figini, Marco Giustetto, and Paul Arne Østvær. In preparation.
9. *Algebraic K-theory of Stiefel varieties*, joint with Sebastian Gant. In preparation.
8. *Motivic K-theory cooperations over the rationals*, joint with Sarah Petersen and Liz Tatum. In preparation.
7. *A $\mathbb{C}$ -motivic w_1 -self-map of periodicity 1*. In preparation.
6. *Periodic phenomena in stable motivic homotopy theory*. Submitted. Available at <https://arxiv.org/abs/2607.18165>
5. *Immersion of C_2 -projective spaces via $K\mathbb{R}$ -theory*, joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang. Submitted. Available at <https://arxiv.org/abs/2604.25260>.
4. *Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras*, joint with Sarah Petersen and Elizabeth Tatum. Submitted. Available at <https://arxiv.org/abs/2509.19542>.
3. *Rings of cooperations for hermitian K-theory over finite fields*. To appear in **Algebraic & Geometric Topology**. Available at <https://arxiv.org/abs/2509.02786>.
2. *On the ring of cooperations for real hermitian K-theory*. To appear in **Annals of K-theory**. Available at <https://arxiv.org/abs/2506.16672>.
1. *Toric Double Determinantal Varieties*, joint with Alexander Blose, Patricia Klein, and Owen McGrath. **Communications in Algebra** 49 (2020), 7, 3085-3093.

TEACHING EXPERIENCE

Instructor of Record

Vanderbilt University

Math 2410 – Methods of linear algebra

Spring 2027

Math 2610/5610 – Ordinary differential equations

Spring 2027

University of Washington

Math 480 – Homotopy theory

Spring 2026

Math 208 – Linear algebra

Summer 2023

Math 208 – Linear algebra

Winter 2023

Graduate Assistant

Electronic Computational Homotopy Theory (eCHT)

Research Seminar

2025–2026

Quadratic Curve Counting

Fall 2024

Park City Math Institute (PCMI)

Experimental Math Lab

Summer 2024

University of Washington

2021–2026

Courses included: Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology

Advising/Mentoring

Directed Reading Mentor

UW

Topics included: Homotopy theory, Representation theory, Geometric group theory, Homological algebra

2022–2025

Advising/Mentorship

Jason Yeung (University of Toronto)

SLMath

Program associate mentor during thematic program

Fall 2026

Spencer Catron (University of Washington)

UW

Graduate student mentor during first year

2023–2024

Math Olympiad Judge

UW Math Hour

2023–2024

Judge for elementary school math competition

HONORS AND AWARDS

Lee and Cozette McFarlan Travel Fellowship

Summer 2025

University of Washington

\$750

For travel to European Talbot in Kolding, DK.

Research Assistantship

Summer 2025

University of Washington

For good research progress.

JC Eaves Scholarship

2020

University of Kentucky

\$2000

For continued excellence in mathematics as a Kentuckian.

Carolyn Bunyan Award

2020

University of Kentucky

\$2000

For mathematics majors interested in continuing education in mathematics.

SERVICE AND PROFESSIONAL ACTIVITIES

Refereeing and Reviewing

Algebraic and Geometric Topology ZBMath

2025-Present

2026-Present

Seminar Organization

Learning Seminar

SLMath

Motivic stable stems and Galois approximations of cellular motivic categories

Fall 2026

DUBTOP Seminar

UW

Topics included: Chromatic homotopy theory, stacks, Goodwillie calculus

2024-2025

Back-To-School seminar

UW

Student lectures on fun math learned over the summer

Fall 2024

Reading seminars

UW

Topics included: Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categories

2023-2024

SELECTED PRESENTATIONS & SEMINAR TALKS

Conference and Colloquium Talks

AMS Sectional: Computational Homotopy Theory

Tempe, AZ

$\mathbb{C}$ -motivic periodic self-maps

November 2026

Motivic Homotopy Theory: Connections and Applications

SLMath, Berkeley, CA

Motivic K-theory cooperations over the rationals

October 2026

Young Topologists Meeting

Copenhagen, DK

Higher Witt K-theories

July 2026

Cascade Topology Seminar

Portland, OR

Higher Witt K-theories

May 2026

Joint Mathematics Meetings 2026

Washington, D.C.

Splittings and cooperations in motivic homotopy theory

January 2026

Graduate Student Geometry & Topology Seminar

Bloomington, IN

On the ring of cooperations for real hermitian K-theory

April 2025

Reed Mathematics Colloquium

Portland, OR

Finding Fixed Points with Invariants

February 2025

Joint Mathematics Meetings 2025

Seattle, WA

On the ring of cooperations for real hermitian K-theory

January 2025

Seminar Talks

New Mexico State University Topology Seminar

Las Cruces, NM

$\mathbb{C}$ -motivic periodic self-maps

September 2026

eCHT Research Seminar

Online

$\mathbb{C}$ -motivic periodic self-maps

September 2026

University of Washington Topology Seminar

Seattle, WA

Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras

February 2026

University of Illinois Urbana-Champaign Topology Seminar

Champaign, IL

Splittings and cooperations in motivic homotopy theory

January 2026

University of Virginia Topology Seminar

Charlottesville, VA

Splittings and cooperations in motivic homotopy theory

November 2025

University of Kentucky Topology Seminar

Lexington, KY

Splittings and cooperations in motivic homotopy theory

November 2025

Duke University Geometry and Topology Seminar

Durham, NC

Splittings and cooperations in motivic homotopy theory

October 2025

University of Colorado Boulder Homotopy Theory Seminar

Boulder, CO

Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras

September 2025

University of Michigan Geometry Seminar

Ann Arbor, MI

On the ring of cooperations for real hermitian K-theory

September 2025

University of Notre Dame Topology Seminar
Splittings and cooperations in motivic homotopy theory
University of Washington Topology Seminar
On the ring of cooperations for real hermitian K-theory

South Bend, IN
 September 2025
 Seattle, WA
 June 2025

Contributed Talks

Reading Seminars

An upper bound on the degree of symmetry of an exotic sphere
Higher Adams differentials and hidden extensions
Motivic twisted K-theory
Higher real K-theories, redshift, and blueshift
Filtered objects, spectral sequences, and t-structures
(A synthetic approach to detecting) v_1 -periodic families
Algebraic and Milnor K-theory
Galois descent and the Picard group of K-theory
The Adams spectral sequence and Hopf algebroids
Symplectic orientations

March 2026
 November 2025
 April 2024
 May 2026
 January 2026
 November 2025
 October 2025
 May 2025
 February 2025
 August 2024

Workshops

$\pi_* THH(\mathbb{Z}_p^{fil})/p$ and $\pi_* THH(\ell_p^{fil})/(p, v_1)$ (Talbot, Nacogdoches TX)
 Hochschild Homology (European Autumn School in Topology, Utrecht, NL)

August 2024
 September 2024